

ITA 0608 - MACHINE LEARNING

2 . For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples.

Aim

To implement and demonstrate the Candidate-Elimination Algorithm using a training dataset stored in a CSV file and obtain the set of all hypotheses consistent with the training examples.

Algorithm

1. Read the training data from the CSV file.
2. Initialize the Specific hypothesis (S) using the first positive example.
3. Initialize the General hypothesis (G) with the most general hypothesis.
4. For each training example:
 - If the example is positive:
 - Generalize S where necessary.
 - Remove inconsistent hypotheses from G.
 - If the example is negative:
 - Specialize G to exclude the negative example while remaining consistent with S.
5. Display the final Specific hypothesis (S) and General hypothesis (G).

Code

```
import pandas as pd
import numpy as np
data = pd.read_csv("training_data.csv")
concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])
specific = concepts[0].copy()
general = [["?" for _ in range(len(specific))] for _ in range(len(specific))]
for i, h in enumerate(concepts):
    if target[i] == "Yes":
        for x in range(len(specific)):
            if h[x] != specific[x]:
                specific[x] = "?"
                general[x][x] = "?"
    else:
        for x in range(len(specific)):
            if h[x] != specific[x]:
                general[x][x] = specific[x]
            else:
                general[x][x] = "?"
general = [g for g in general if g != ["?" for _ in range(len(specific))]]
print("Specific Hypothesis:")
print(specific)

print("\nGeneral Hypothesis:")
for g in general:
    print(g)
```

Output

```
Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
```

Result

The Candidate-Elimination Algorithm was successfully implemented using a CSV file. The algorithm generated the Specific Hypothesis (S) and the General Hypothesis (G), representing the version space of all hypotheses consistent with the given training examples.